Lecture Notes

Knowledge Graph Enhanced Educational Notes

1. Data

Definition:
Are simply a value are set of values of different type which is called data types like string, integer, char etc. or collection of information.

2. Structure (Concept)

Definition:
Way of organizing information, so that it is easier to use.

Visual Representation:

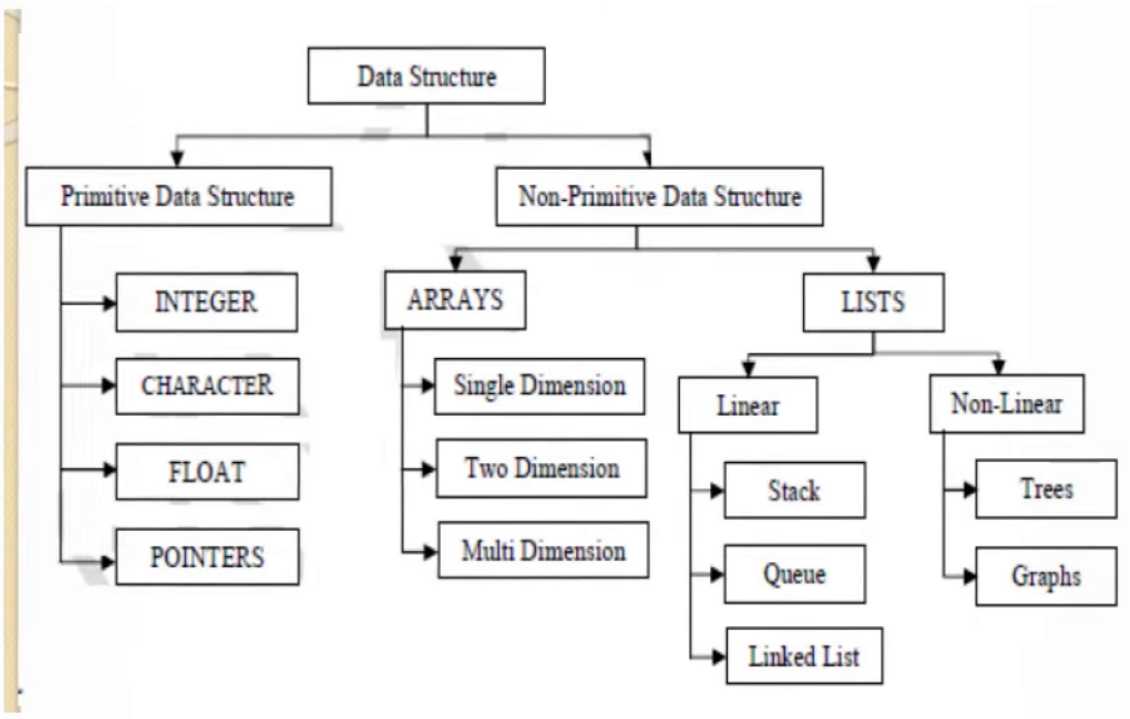

Figure 2: Diagram illustrating: Way of organizing information, so that it is easier to use.

3. Data Objects

Definition:
A unique location in computer memory that contains the values.

4. Program

Definition:
Program is a key concept in this domain. It includes components such as Algorithm, Data Structure.

- Components and Structure

- Algorithm: Step by Step procedure to solve a particular function.
- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.

5. Efficiency

Definition:
Allows data to be used efficiently.

6. Element Relationship

Definition:
Consideration of elements stored and their relationship to each other.

7. Data Manipulation

Definition:
A scheme for organizing related pieces of information that can be manipulated.

- Related Concepts
- Data Structure Operations: Commonly used operations performed on data structure.

8. Why Data Structure

Definition:
Necessity to solve complex requirements in an efficient way.

- Related Concepts
- Complex Human Requirements: Human requirement with computer are going to complex day by day.
- Fastest Solutions: Provides fastest solution of human requirements.
- Efficient Solutions for Complex Problems: Provides efficient solution of complex problem.
- Space Complexity Solution: Provides solution for Space Complexity.
- Time Complexity Solution: Provides solution for Time Complexity.

9. Complex Human Requirements

Definition:
Human requirement with computer are going to complex day by day.

10. Fastest Solutions

Definition:
Provides fastest solution of human requirements.

11. Efficient Solutions For Complex Problems

Definition:
Provides efficient solution of complex problem.

12. Logical/Mathematical Description

Definition:
Logical or mathematical description of the structure.

- Related Concepts

- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.

13. Structure Implementation

Definition:

Implementation of structure on a computer using different computer languages.

- Related Concepts

- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.
- C++ Language: A computer language mentioned for implementation purpose in the syllabus.

14. C++ Language

Definition:

A computer language mentioned for implementation purpose in the syllabus.

15. Quantitative Analysis

Definition:

Determining the amount of memory needed to store the structure and the time required to process the structure.

- Related Concepts

- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.

16. Data Structure Classification

Definition:

Categorization of data structures into types.

16.1 Non-Primitive Data Structure

Data structures that are derived from the primitive data structures.

- Homogeneous/Heterogeneous Grouping: Structuring of a group of homogeneous (same type) or heterogeneous (different type) data items.
- Linear Data Structure: A data structure that has homogeneous elements and where elements are in a sequence.
- Non-Linear Data Structure: A data structure in which data item is connected to several other data items.

Other Components:

- Primitive Data Structure: Data structures that are directly operated upon the machine-level instructions.

- Related Concepts

- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.

17. Built-In/Predefined Data Types

Definition:

Standard / Predefine/ Built data types in of any one of the computer languages.

18. Elements In A Sequence

Definition:
Elements are in a sequence and form a liner series.

Visual Representation:

Figure 18: Diagram illustrating: Elements are in a sequence and form a liner series.

19. Hierarchical/Parent-Child Relationship

Definition:
May exhibit either a hierarchical relationship or parent child relationship.

20. Static Data Structure

Definition:
The size of the structure is fixed.

- Related Concepts
- Array: A non-primitive data structure that includes single dimensional and multi-dimensional types, and is an example of static and linear data structures.

21. Dynamic Data Structure

Definition:
The size of the structure in not fixed and can be modified during the operations performed on it.

- Related Concepts
- Linked List: A non-primitive data structure, an example of dynamic and linear data structures.

22. Data Structure Operations

Definition:
Commonly used operations performed on data structure.

- Components and Structure

- Traversing: Processing each element in the list.
- Search: Finding the location of the element with a given value or the Selection record with a given key.
- Insertion: Adding a new element to the list.
- Deletion: Removing an elements from the list.
- Sorting: Arranging the elements in some type of order.
- Merging: Combining two list into a single list.

- Related Concepts

- Data Structure: A particular way of organizing data in a computer so that it can be used efficiently.

23. Array

Definition:

A non-primitive data structure that includes single dimensional and multi-dimensional types, and is an example of static and linear data structures.

24. Linked List

Definition:

A non-primitive data structure, an example of dynamic and linear data structures.

25. Stack

Definition:

An example of a linear data structure.

26. Queue

Definition:

An example of a linear data structure.

27. Tree

Definition:

An example of a non-linear data structure.

Visual Representation:

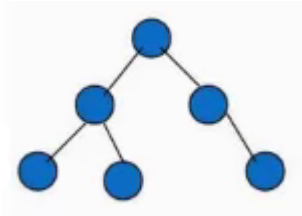

Figure 27: Diagram illustrating: An example of a non-linear data structure.

28. Graph

Definition:

An example of a non-linear data structure.

Visual Representation:

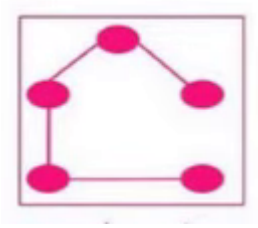

Figure 28: Diagram illustrating: An example of a non-linear data structure.

29. Hash Table

Definition:
An example of a non-linear data structure.

30. Integer

Definition:
A type of primitive data structure.

Visual Representation:

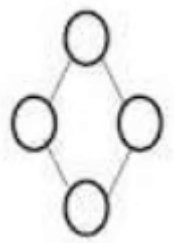

Figure 30: Diagram illustrating: A type of primitive data structure.

31. Character

Definition:
A type of primitive data structure.

32. Float

Definition:
A type of primitive data structure.

33. Pointers

Definition:
A type of primitive data structure.

34. Data Structures (Subject)

Definition:
a new subject that is data structures.

- Related Concepts

- Data (Core Concept): a prerequisite before understanding what data structure is.

- Data Structures Importance: the reason why companies are preferring this subject over other subjects, as it is very important and useful in day-to-day life and used more frequently.

35. Data (Core Concept)

Definition:

a prerequisite before understanding what data structure is.

- Components and Structure

- Data vs. Information Distinction: a distinction between data and an information, which is a very confusing topic.

- Related Concepts

- Data (Dictionary Definition): quantities, characters or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical or mechanical recording media.

36. Data (Dictionary Definition)

Definition:

quantities, characters or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical or mechanical recording media.

- Related Concepts

- Operations on Data: operations performed by a computer on quantities, characters or symbols.
- Storage and Transmission of Data: data may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical or mechanical recording media.

Visual Representation:

Figure 36: Diagram illustrating: Are simply a value are set of values of different type which is called data types like string, integer, char etc. or collection of information.

37. Operations On Data

Definition:

operations performed by a computer on quantities, characters or symbols.

38. Storage And Transmission Of Data

Definition:

data may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical or mechanical

recording media.

39. Unprocessed Data (Example)

Definition:
just a collection of characters or symbols where one won't be able to understand it properly.

- Related Concepts
- Processed Data: data that has been manipulated, such as reversing a whole string, to make it understandable.
- Systematic Data Arrangement: arranging data in a systematic way so that it gets a structure and becomes meaningful.

40. Processed Data

Definition:
data that has been manipulated, such as reversing a whole string, to make it understandable.

- Components and Structure
- Information: meaningful or processed data.

41. Meaningful Information (Goal)

Definition:
information from which some meaning can be extracted, making it quite understandable.

42. Data Management

Definition:
managing data in such a way so that it can produce some meaningful information.

- Related Concepts
- Meaningful Information (Goal): information from which some meaning can be extracted, making it quite understandable.
- Data Structure (Definition): the systematic way to organize data so that it can be used efficiently.

43. Data Structure (Definition)

Definition:
the systematic way to organize data so that it can be used efficiently.

- Components and Structure
- Systematic Data Arrangement: arranging data in a systematic way so that it gets a structure and becomes meaningful.
- Arrays (Example Data Structure): a classical data structure, thought of as a sequence or collection of some symbols, characters, integers, or floats.
- Stack Data Structure: a data structure used in implementing the redo and undo feature.
- Two-Dimensional Array: a data structure (e.g., 37x40) used to store pixel information (each pixel's color) for bitmap images.
- Graph Data Structure: a data structure used to store friendship information on social networking sites.

- Related Concepts
- Efficiency (Time/Space): using data efficiently in terms of time as well as space.

44. Efficiency (Time/Space)

Definition:
using data efficiently in terms of time as well as space.

45. Undo/Redo Feature

Definition:
a famous feature in applications like Google Docs, PowerPoint, or Microsoft Word that uses stack data structures.

- Related Concepts
- Undo Stack: a stack where each information entered gets stored; elements are popped out when control z (undo) is pressed.
- Redo Stack: a stack where elements popped from the undo stack are placed; used when control y (redo) is pressed to get the operation again.

46. Undo Stack

Definition:
a stack where each information entered gets stored; elements are popped out when control z (undo) is pressed.

- Related Concepts
- Redo Stack: a stack where elements popped from the undo stack are placed; used when control y (redo) is pressed to get the operation again.

47. Redo Stack

Definition:
a stack where elements popped from the undo stack are placed; used when control y (redo) is pressed to get the operation again.

48. Bitmap Images

Definition:
images stored as a series of tiny dots called pixels.

- Components and Structure
- Pixels: tiny dots or small squares, each assigned a color and then arranged in a pattern to form an image.

Visual Representation:

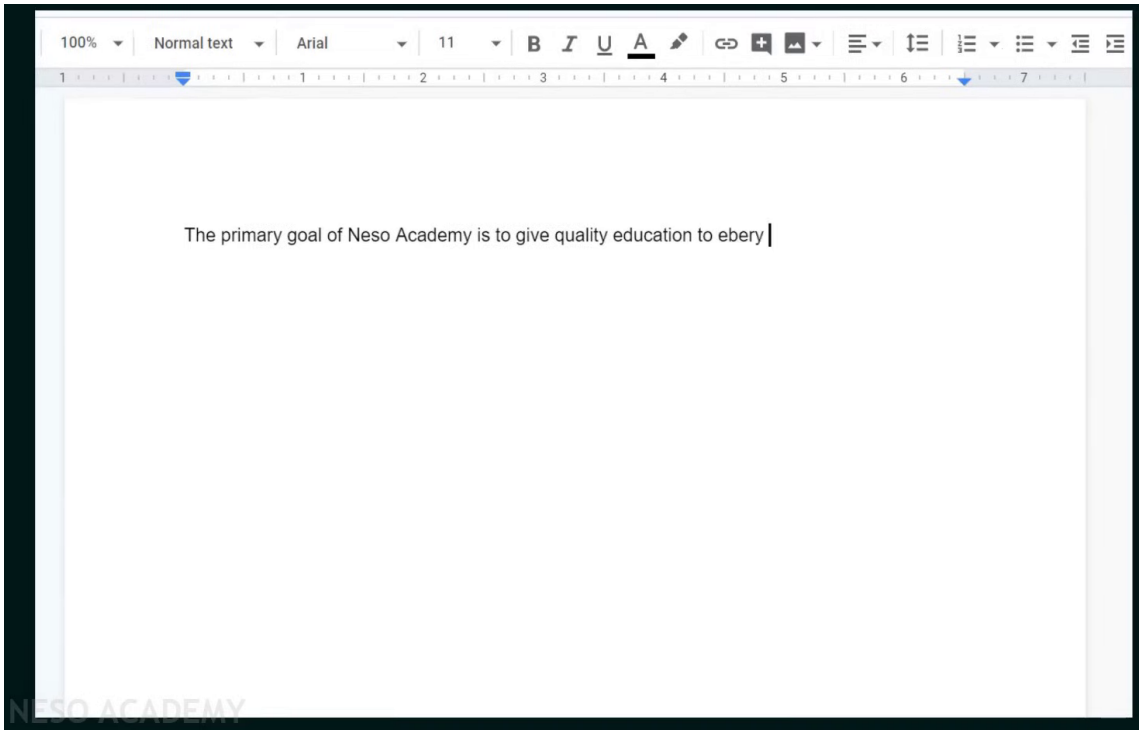

Figure 48: Diagram illustrating: images stored as a series of tiny dots called pixels.

49. Graph Nodes

Definition:
individuals (e.g., James, Mark, Leah, Lucy) represented in a graph data structure.

50. Graph Edges

Definition:
connections or relationships between individuals (nodes) represented in a graph data structure.

51. Data Structures Importance

Definition:
the reason why companies are preferring this subject over other subjects, as it is very important and useful in day-to-day life and used more frequently.